

Symmetry as a resource for parallel computation in nonlinear dynamical systems

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Abstract

Symmetry is usually used to simplify the analysis of physical systems. Here, we show that it can also be used as a resource for computation. We introduce symmetry-induced parallelism, a mechanism by which a nonlinear dynamical system represents and progresses several computational states within a single physical state. Without separating tasks in space or time, this mechanism relies on symmetry to bind compatible computational states together, so that one relaxation process converges to a configuration encoding multiple outcomes. We demonstrate this mechanism in relaxed spin networks using Boolean gates and adders as examples. We show that every vectorial Boolean function admits, at least, twofold simultaneous evaluations, giving a universal minimal instance of the mechanism. Structured tasks can support larger sets; for example, an N -bit adders support up to $2N+2$ additions. These results establish symmetry-induced parallelism as a route to parallel computation in nonlinear classical systems.

Keywords: dynamical computations, combinatorial optimization, Ising machines, parallel computing, quantum-inspired-computing

24 1 Introduction

25 The notion of computational parallelism is commonly associated with computational
26 effort performed by dedicated entities separated in space or time. An everyday example
27 is a multicore processor, in which different cores—physically distinct collections of
28 logic gates and other components—perform their own computational tasks, either
29 independently or in communication with one another. A notable exception is quantum
30 parallelism, which employs superpositions of states in the tensor-product space of
31 individual qubits (see, e.g., Section 1.4.2 in [1]). To a certain extent, this approach
32 can be mimicked in classical systems that admit superpositions of states, for example,
33 polarizations of the electromagnetic field. However, to employ superpositions, the
34 system must be linear and hence have limited computational capabilities [2–5]. This
35 is exacerbated by the fact that reproducing the structure of the tensor product, which
36 alleviates limitations of linear systems in quantum computing, is challenging to scale
37 in classical systems [6, 7].

38 We show that multiple simultaneous computations can be enabled by the principle
39 of symmetry, which does not pose intrinsic limitations on dynamics. The foundational
40 idea of symmetry-induced computational parallelism is based on the property that,
41 in a system with continuous symmetries — governed by equations of motion that are
42 invariant under a set of transformations — the terminal state is a representative of
43 the entire manifold of states covered by the symmetry transformation. In turn, the
44 relation between the “physical” states, i.e., points in the system’s phase space, and
45 the “computational” states represented by them needs not obey the same symmetry
46 as dynamics. Consequently, the symmetry transformation may traverse multiple com-
47 putational states, signifying the presence of multiple evaluations in the final state.

48 This mechanism can be illustrated by the following diagram

$$\begin{array}{ccc}
 \psi(\infty|\psi(0;\Sigma_0)) & \xrightarrow{F} & f(\Sigma_0) \\
 \mathcal{T}_r \downarrow & & \downarrow \mathcal{T}_r \\
 \psi(\infty|\mathcal{T}_r[\psi(0;\Sigma_0)]) & \xrightarrow{F} & f(\mathcal{T}_r\Sigma_0)
 \end{array} \tag{1}$$

49 Here, ψ is the state of the dynamical system, and $\psi(\infty|\psi(0))$ denotes the terminal
 50 state of the system evolving from the initial state $\psi(0)$. Next, $\psi(0;\Sigma_0)$ denotes the
 51 initial state containing the encoded computational input state Σ_0 . The mapping F
 52 maps the state of the dynamical system to a computational state. The property that
 53 the evolution represents an evaluation of the function f is depicted by the upper line in
 54 the diagram, which relates $\psi(\infty|\psi(0;\Sigma_0))$ to $f(\Sigma_0)$. The symmetry transformation
 55 is denoted by $\mathcal{T}_r$, where r is the transformation parameter. Although diagram (1) does
 56 not presume any particular structure of the symmetry group, in view of the particular
 57 dynamical realization of symmetry-induced parallelism, we consider a one-parameter
 58 group of transformation.

59 The invariance of the dynamical equations transfers the action of the symme-
 60 try transformation to the initial state: $\mathcal{T}_r[\psi(\infty|\psi(0;\Sigma_0))] = \psi(\infty|\mathcal{T}_r[\psi(0;\Sigma_0)])$. In
 61 turn, the transformation of the initial state $\psi(0)$ may result in a variation of the
 62 input computational state represented by $\psi(0)$. Finally, the right vertical arrow in
 63 Eq. (1) defines the effective action of the symmetry transformation on the evaluation
 64 outcome; it effectively contains the action of the evaluated function f on a (possibly
 65 nontrivial) set of pairwise distinct input states, $\mathcal{S}^{(\text{in})} = \{\Sigma_0, \Sigma_{r_1}, \Sigma_{r_2}, \dots\}$, reached
 66 through the symmetry transformations. Here, $\Sigma_{r_1} = \mathcal{T}_{r_1}\Sigma_0$ is the input state reached
 67 at $r = r_1$, $\Sigma_{r_2} = \mathcal{T}_{r_2}\Sigma_0$ is the input state reached at $r = r_2$, and so on.

68 Diagram (1) emphasizes the property of the final dynamical state to contain within
 69 itself the results of processing multiple input states from the set $\mathcal{S}$. Of course, this
 70 diagram itself does not guarantee that set $\mathcal{S}$ is nontrivial and contains any other state

71 besides Σ_0 . The main result of the present paper is the demonstration of symmetry-
72 induced parallelism in a concrete dynamical system: a relaxation-based dynamical
73 Ising machine (R-DIM) driven by the V_2 model [8].

74 Usually, dynamical Ising machines [9, 10], which search for the ground state of the
75 classical Ising model, comprise bistable dynamical elements, so that the machine's ter-
76 minal state, with continuous dynamical variables grouping around two values, say, ± 1
77 in the respective units, can be immediately interpreted as a classical spin configura-
78 tion. The R-DIM abandons the requirement of convergence to a spin-like state in favor
79 of predictable computational capabilities. As a result, the R-DIM's terminal states are
80 not necessary spin-like and may require postprocessing to extract a feasible spin con-
81 figuration. However, the structure of the terminal states of the V_2 model makes such
82 postprocessing trivial even when the terminal state is nonbinary and there is ambi-
83 guity in the correspondence between the machine's state and the spin configuration.
84 This property plays a key role in enabling symmetry-induced parallelism.

85 The principle of symmetry-induced parallelism is applicable in the general infor-
86 mation processing context. However, in the present paper, we limit ourselves to the
87 evaluation of Boolean functions as a representative computational task.

88 The presented results address three aspects of symmetry-induced parallelism.
89 First, we show that relaxed spin configurations can represent multiple binary states,
90 providing a concrete physical witness of the general symmetry principle. Second,
91 we relate consistent simultaneous computations to the isotonicity of the evaluated
92 Boolean function along the respective chain of input states. This yields a universal
93 two-state admissibility result for arbitrary vectorial Boolean functions and stronger
94 multi-state examples for structured functions. Third, we study convergence of the
95 dynamics. We prove almost-sure convergence for a class of simple Boolean func-
96 tions and demonstrate numerically symmetry-induced parallelism for N -bit adder,

97 while showing that convergence probability depends strongly on the chosen spin
 98 representation of the Boolean function.

99 2 Results

100 2.1 V_2 -supported symmetry-induced parallelism

101 We start with considering in detail symmetry-induced parallelism for the example of
 102 the AND gate. Symmetry-induced parallelism is enabled by employing the continuous
 103 degree of freedom of the relaxed spins for storing information. The spins representing
 104 the input arguments are set at fixed displaced positions on the phase circle, as illus-
 105 trated by Fig. 1(a). If the relaxed spin system ends up in a proper state, for example,
 106 as shown in Fig. 1(a) by the split arrow, the resultant state of relaxed spins, includ-
 107 ing the auxiliary and input ones, represents a series of binary states corresponding
 108 to evaluating AND and OR for various input arguments. Thus, in the case shown
 109 in Fig. 1(a), the single evaluation of the relaxed spin terminal state produced the
 110 results of six computations. These results can be read-off by applying the respective
 111 symmetry transformation, for instance, by rotating the phase circle, as illustrated by
 112 Fig. 1(b). We note that the transformations required for extracting the results are
 113 fully determined by the arrangements of the fixed spins on the phase circle.

114 An arrangement of relaxed spins along the phase circle defines the order in
 115 traversing various sets of function arguments. A convenient framework for describing
 116 transformations occurring during single full rotation of the phase circle is that of par-
 117 tially ordered sets (standard background on posets, chains, and other related concepts
 118 can be found in [11]). The power set of the set of indices enumerating relaxed spins
 119 representing the input parameters is ordered by inclusion. A *chain* $\mathcal{J} : \emptyset \subset \{m\} \subset \dots$
 120 defines a sequence of spin-flip transformations. For the shown example, initially, none
 121 of the spins are inverted compared to the network terminal state, at the first step, the
 122 m -th spin is inverted, $\sigma_m \rightarrow -\sigma_m$, and so on. Applying this sequence to a binary state

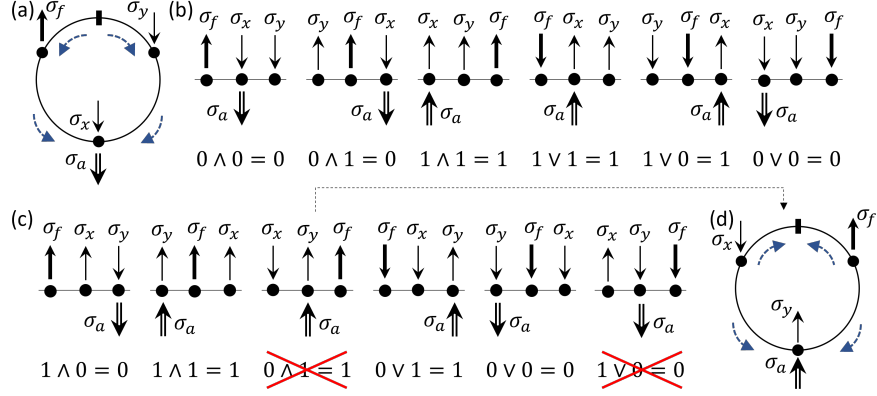


Fig. 1 Symmetry-induced parallelism in the AND/OR gate. The bold and split arrows show the auxiliary spin and the output relaxed spin, respectively. Bold dots indicate clusters in the stable equilibrium state: two clusters are trivial, and one cluster contains two relaxed spins. (a) An example of arrangement of the input spins and the solution delivered by the V_2 model corresponding to the starting state in chain shown in (b). Curved dashed lines indicate $\hat{X}_a$ of the relaxed spin $\psi_a = (\sigma_a, \hat{X}_a)$ placed inside the respective intervals with $\sigma_a = 1$. (b) An example of a maximal chain encoded in the relaxed spin state shown in (a). (c) An example of broken chain: the only chain, out of 11 different nontrivial chains that are not related to each other by the symmetries of AND and OR functions and the phase circle, inconsistent with the gate functionality. (d) The stability diagram of the state yielding the broken chain.

123 produces a chain of binary states. Thus, a terminal state of the V_2 model plays the
 124 role of the basepoint of the chain of states yielded by the symmetry transformation.
 125 Notably, this sequence of states, $\mathcal{S} : \sigma(0) \prec \sigma(1) \prec \dots$, forms chains while restricted
 126 to both spins representing the function arguments, $\sigma^{(\text{in})}(0) \prec \sigma^{(\text{in})}(1) \prec \dots$, and spins
 127 representing the function output, $\sigma^{(\text{out})}(0) \prec \sigma^{(\text{out})}(1) \prec \dots$

128 The same property can be approached from the perspective of the Boolean func-
 129 tions, $f : \{-1, 1\}^N \rightarrow \{-1, 1\}^M$, itself. A chain of bit-flip transformations $\mathcal{J}^{(\text{in})} :$
 130 $\emptyset = J_0 \subset J_1 \subset \dots$ applied to the vector of the function arguments $\sigma^{(\text{in})}$ produces a
 131 chain of arguments $\mathcal{J}^{(\text{in})} : \sigma^{(\text{in})} = \sigma^{(\text{in})}(0) \prec \sigma^{(\text{in})}(1) \prec \dots$. We say that function f
 132 is $\mathcal{J}^{(\text{in})}$ -isotone [12] if $f(\sigma^{(\text{in})}(0)) \preceq f(\sigma^{(\text{in})}(1)) \preceq \dots$, with the order transported
 133 from the respective bit-flip chain $\mathcal{J}^{(\text{out})}$ in $\{-1, 1\}^M$.

134 These two perspectives are connected by observing that during the symmetry
 135 transformation, each spin inverts at most once. Then, there exists such an associa-
 136 tion between components of $\sigma^{(\text{out})}$ and $\sigma^{(\text{in})}$ (we will call it *consistent* with f) that

the symmetry transformation produces such bit-flip chains, $\mathcal{S}^{(\text{in})}$ and $\mathcal{S}^{(\text{out})}$, that $\sigma^{(\text{out})}(0) = f(\sigma^{(\text{in})}(0))$, $\sigma^{(\text{out})}(1) = f(\sigma^{(\text{in})}(1))$, and so on. Figure 1(a) shows an example of such an association for the AND/OR function.

Using this framework, we show that, at least for simple functions, spin networks driven by the V_2 model converge to states consistent with symmetry-induced parallelism. Let $f : \{-1, 1\}^N \rightarrow \{-1, 1\}$, where N includes the fixed auxiliary spins, if needed, be a Boolean function whose spin representation does not involve additional internal relaxed spins. We denote this class of Boolean functions by $\mathcal{F}_1(N)$, where 1 stands for the number of dynamical degrees of freedom. For example, $\mathcal{F}_1(N)$ includes, for $N = 3$, AND/OR and NAND/NOR, but does not include XOR [14].

Theorem 1 *Let the relaxed spins representing arguments of function $f \in \mathcal{F}_1(N)$ be arranged in such a way that the symmetry transformation produces a chain $\mathcal{S} : \sigma^{(\text{in})} = \sigma^{(\text{in})}(0) \prec \sigma^{(\text{in})}(1) \prec \dots \prec \sigma^{(\text{in})}(S-1)$, with $1 \leq S \leq N+1$, and f be $\mathcal{S}$ -isotone. Then, the state of the output relaxed spin consistent with f is the only stable equilibrium of the network.*

Proof Without loss of generality, we enumerate the input relaxed spins in the order of traversing the phase circle counterclockwise: $X_m \leq X_n$, if $1 \leq m < n \leq N$. Next, we consider the sign of $\dot{X}_a$ when the dynamical relaxed spin ψ_a is placed inside different S intervals formed by the boundaries $\{\pm 1\}$ and positions of the fixed relaxed spins (see, e.g., Fig. 1(a)). In the outmost right, S -th, interval, $X_a = X^{(S)}$ with $X_N < X^{(S)} < 1$, we have

$$\text{sgn} \left[\dot{X}_a \left(X^{(S)} \right) \right] = \sigma_a \text{sgn} \left[F \left(\sigma^{(\text{in})} \right) \right], \quad (2)$$

where $F \left(\sigma^{(\text{in})} \right) = \sum_m A_{a,m} \sigma_m^{(\text{in})}$ depends only on the input spins. The necessary condition of maximum cut is $\sigma_a F \left(\sigma^{(\text{in})} \right) \leq 0$, signifying that inverting σ_a does not increase the total weight of cut edges incident to a . Since maximizing cut determines σ_a unambiguously, we presume that $\sigma_a F \left(\sigma^{(\text{in})} \right)$ is strictly negative. Thus, $\text{sgn} [F(\sigma)]$ is negation of the represented function, $\text{sgn} [F(\sigma)] = \neg f(\sigma)$.

Thus, when the spin configuration $(\sigma^{(\text{in})}, \sigma_a)$ satisfies the function truth table (yields the maximum cut), the free relaxed spin ψ_a goes away from the right boundary of $[-1, 1]$. Since the equations of motion are antisymmetric with respect to inversion of the phase circle, in the outmost left interval, $X_a = X^{(1)}$ with $-1 < X^{(1)} < X_1$, we have $\text{sgn}[\dot{X}_a(X^{(1)})] = -\text{sgn}[\dot{X}_a(X^{(S)})]$. Hence, the state yielding the output of the Boolean function is dynamically stable and attracting, with $X_a(\infty) \in [X_1, X_N]$.

Next, we notice that the sign of $\dot{X}_a$ in an “internal” interval is the same as when this interval becomes outmost right after the symmetry transformation:

$$\text{sgn}[\dot{X}_a(X^{(S-k)})] = \sigma_a \text{sgn}[F(\sigma^{(\text{in})}(k))]. \quad (3)$$

This follows from the symmetry of the equations of motion, or can be directly checked by tracing how the sign of the right-hand-side of the equations of motion changes while crossing from one interval to another. Since $\text{sgn}[F(\sigma)]$ is $\mathcal{S}$ -isotone, it changes sign exactly once while traversing the intervals. The boundary between the intervals where the sign changes is the stable equilibrium of the V_2 model, and this is the sole stable equilibrium. $\square$

Thus, for almost every initial condition of the relaxed spin network, the dynamics converges to the terminal state representing the consistent chain of output evaluations. The exceptional set consists of initial conditions lying on unstable invariant sets.

When the represented function is not isotone along the symmetry-induced bit-flip chain, we will call the chain *broken*. For the AND/OR gate, such case is shown in Fig. 1(c). Following the similar argument as above, one can see that, broken chains manifest in (and, in fact, are equivalent to) the existence of multiple stable equilibria. For example, for the first state in Fig. 1(c), another stable equilibrium is when ψ_a clusters with ψ_f . Settling in different equilibria determines different variations of the output parameter with the symmetry transformation, which cannot be consistent with the function output.

Function f being isotone along a bit-flip chain is the property of the chain. Even if the function is not isotone along a saturated chain (with only one bit inverted at

each step along the chain), the function may be isotone along coarser chains. As an ultimate example, any function is isotone along bit-flip chains with less than three elements.

Proposition 1 (Universal twofold admissibility) *Let $f : \{-1, 1\}^N \rightarrow \{-1, 1\}^M$ be a vectorial Boolean function, and the input variables be split as $\sigma = (\sigma^{(A)}, \sigma^{(B)})$, represented by relaxed spins placed on the phase circle as two clusters with distinct continuous coordinates $X^{(A)}$ and $X^{(B)}$. For every input $(\sigma^{(A)}, \sigma^{(B)})$, the two-element sequence induced by the symmetry transformation $\mathcal{S}^{(\text{in})} : (\sigma^{(A)}, \sigma^{(B)}) \prec (\sigma^{(A)}, -\sigma^{(B)})$ defines a chain, and f is $\mathcal{S}^{(\text{in})}$ -isotone. Hence every vectorial Boolean function admits at least one twofold symmetry-induced parallel evaluation for every choice of input partition.*

Indeed, the outputs $\tau = f(\sigma^{(A)}, \sigma^{(B)})$ and $\tau' = f(\sigma^{(A)}, -\sigma^{(B)})$ determine a unique partition of the output coordinates into unchanged and inverted sets,

$$J^{(A)} = \{i : \tau'_i = \tau_i\}, \quad J^{(B)} = \{i : \tau'_i = -\tau_i\}. \quad (4)$$

With this output partition, the pair $\mathcal{S}^{(\text{out})} : f(\sigma^{(A)}, \sigma^{(B)}) \preceq f(\sigma^{(A)}, -\sigma^{(B)})$ is an admissible two-element output chain, enabled by aligning the relaxed spins representing the output bits with coordinates in $J^{(A)}$ and $J^{(B)}$ with the clusters positioned at $X^{(A)}$ and $X^{(B)}$, respectively.

The significance of this proposition is physical rather than combinatorial. The relaxed spin network driven by the V_2 model realizes this admissible chain by driving the output spins in $J^{(A)}$ and $J^{(B)}$ at the required positions on the phase circle, thereby representing the two evaluations in a single terminal state.

For the relaxed spin networks this implies that clustering some of varied parameters may eliminate broken chains. For the AND/OR gate, assigning $\psi_x = (\sigma_x, X^{(\text{in})})$ and

209 $\psi_y = (\sigma_x, X^{(\text{in})})$ makes σ_x and σ_y revert simultaneously with the rotation of the phase
 210 circle thus limiting the maximum number of variations to two.

211 2.2 N -bit adder

212 To demonstrate symmetry-induced parallelism in a more complex setup, we consider
 213 N -bit ripple-carry adder, constructed of N full adders FA, as shown in Fig. 5(b, c), by
 214 identifying the spin representing the carry-out bit in the k -th FA gate with the spin
 215 representing the carry-in bit in the $(k + 1)$ -th FA gate. Alternatively, these bits could
 216 be represented by different nodes connected by an edge with a negative weight.

217 To ensure the isotonicity, we reduce the number of variable parameters by cluster-
 218 ing the input bits x_i and y_i . In a single FA, this corresponds to having two arguments,
 219 c_{in} and (x, y) , that can be varied by the symmetry transformation and two output
 220 parameters, c_{out} and s . Of course, in a single FA, other pairings are possible, however,
 221 in the ripple-carry adder, the relaxed spins representing the carry-in bits in the k -th
 222 FA's, with $k > 1$, are determined dynamically and should not be tied to the input bits.
 223 Thus, the number of simultaneous computations (the length of a maximum chain)
 224 obtained through symmetry-induced parallelism of the N -bit adder is limited from
 225 above by $2N + 2$.

226 Figures 2(a, b) show how symmetry-induced parallelism reproduces the carry-
 227 select effect [13] for N -bit adders. This is a consequence of the universal twofold
 228 admissibility. In this case, the input relaxed spins are clustered together, while
 229 the relaxed spin representing the carry-in bit is displaced on the phase circle. The
 230 spins representing the output sum and the carry-out bit are dynamically distributed
 231 between these two clusters. This is depicted schematically in Fig. 2(a,b) by two vec-
 232 tors: $\sigma^{(\text{s}, \text{out})}(1)$ corresponding to the relaxed spins that cluster with the input spins,
 233 and $\sigma^{(\text{s}, \text{out})}(2)$ representing those clustering with the carry-in spin. The relaxed spin
 234 representing the carry-out bit belongs to one of these two groups, either to $\sigma^{(\text{s}, \text{out})}(1)$

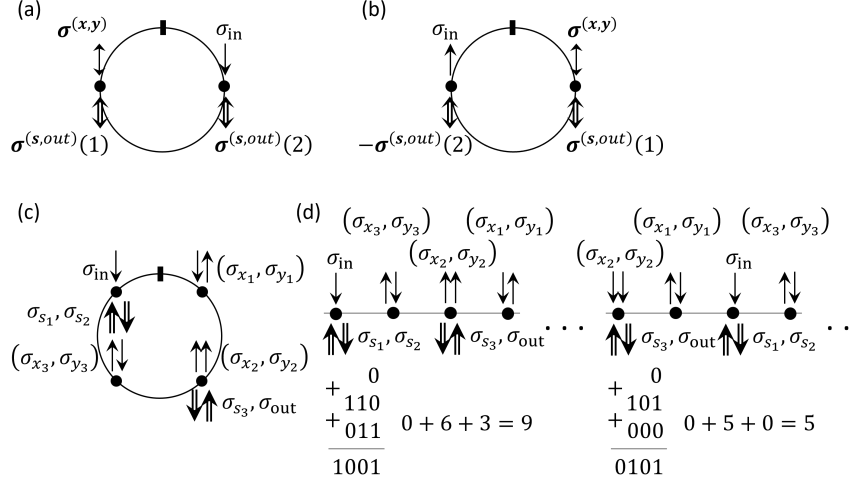


Fig. 2 Structure of states representing multiple outputs for N -bit adder. (a, b) The carry-select effect reproduced by symmetry-induced parallelism: $\sigma^{(x,y)}$ shows that all input relaxed spins are clustered together, $\sigma^{(s,out)}(1)$ and $\sigma^{(s,out)}(2)$ show the output spins (including the carry-out spin) clustered with the input and carry-in spins, respectively. (a) The carry-in bit is reset, $c_{in} = 0$, (b) the carry-in bit is set, $c_{in} = 1$. (c) An example of a state at the start of a maximal chain for a 3-bit adder. (d) The first and the third states in the chain starting from the state shown in (c): $0 + 6 + 3 = 9$, $0 + 7 + 2 = 9$, $0 + 5 + 0 = 5$, $0 + 4 + 1 = 5$, $1 + 4 + 1 = 6$, $1 + 0 + 5 = 6$, $1 + 2 + 7 = 10$, $1 + 6 + 3 = 10$.

or to $\sigma^{(s,out)}(2)$. Depending on the actual value of the carry-in bit, the sum is read either directly from the terminal state ($c_{in} = 0$) or after rotating the phase circle clockwise by π ($c_{in} = 1$) when the spins collected in $\sigma^{(out)}(2)$ change their sign.

From the computer arithmetic perspective, allowing for the carry-in bits in individual FA's to settle dynamically can be regarded as N -bit adder implemented as N 1-bit carry-select adders. In this regard, it must be emphasized that in conventional carry-select adders each computation, assuming a particular value of the carry-in bit, is performed in a dedicated block. Owing to symmetry-induced parallelism, in the relaxed spin realization of N -bit adder, all computations are done within the same system.

Figure 2 shows the first state of a chain for a 3-bit adder. The state represents “ $0 + 6 + 3 = 9$ ”, where 0 corresponds to the carry-in bit. It should be noted that the order of arrangement of the variable inputs matters. Different arrangements may produce different chains that coincide only at 0, 2π (full inversion of the discrete

249 component of the relaxed spin), and 4π (return to the original state) angles of rotation
 250 of the phase circle.

251 While these considerations demonstrate the principal possibility of the system-
 252 induced parallelism in a network driven by the V_2 model, they leave open the
 253 question of convergence to the required state starting from a generic initial state. As
 254 Fig. 3 demonstrates, during the evolution, the system of relaxed spins may encounter
 255 dynamical bottlenecks that may lead the system to a metastable state.

256 Figure 3 shows the probability of obtaining correct results for the example of
 257 adding two 32-bit numbers and the carry-in bit $0 + 3411433493 + 2079581652$. The par-
 258 allel computations were encoded by displacing relaxed spins $\langle 17, 3, 25, 0, 8, 30, 12, 21 \rangle$,
 259 with 0 standing for the relaxed spin representing the carry-in bit, as seen while travers-
 260 ing the phase circle from $X = 1$ to $X = -1$. Thus, this arrangement of input relaxed
 261 spins encoded the following branches

$$\begin{aligned}
 \{\emptyset\} & \quad 0 + 3411433493 + 2079581652 = 5491015145, \\
 \{17\} & \quad 0 + 3411499029 + 2079516116 = 5491015145, \\
 \{3\} & \quad 0 + 3411499025 + 2079516112 = 5491015137, \\
 \{25\} & \quad 0 + 3394721809 + 2062738896 = 5457460705, \\
 \{0\} & \quad 1 + 3394721809 + 2062738896 = 5457460706, \\
 \{8\} & \quad 1 + 3394721937 + 2062738768 = 5457460706, \\
 \{30\} & \quad 1 + 3931592849 + 1525867856 = 5457460706, \\
 \{12\} & \quad 1 + 3931590801 + 1525869904 = 5457460706, \\
 \{21\} & \quad 1 + 3930542225 + 1524821328 = 5455363554,
 \end{aligned} \tag{5}$$

262 where $\{\emptyset\}$ corresponds to the terminal state as seen at the original orientation of the
 263 phase circle, $\{17\}$ denotes the inversion of the 17-th pair of bits in the input numbers,
 264 $\{3\}$ corresponds to additional inversion of the 3-rd pair, and so on. The characteristic

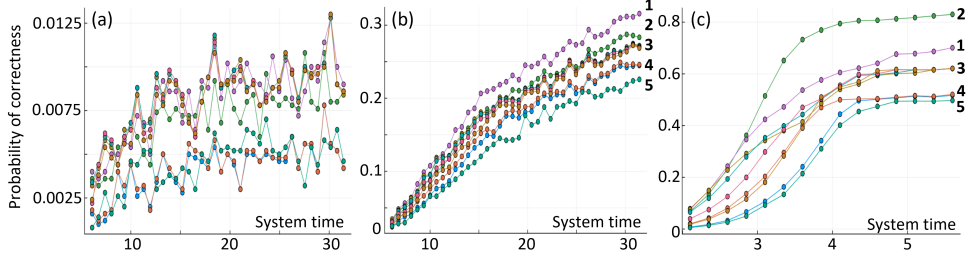


Fig. 3 An example of the probability to obtain correct results for individual branches of adding two 32-bit numbers as a function of duration of evolution in units of the model equations of motion [Eq. (8)]. (a) Concurrent N -bit adder with the weights shown in Fig. 5(b) [14]. (b) Concurrent N -bit adder with the weights updated as described in the text. The numbers in braces in (b) and (c) designate groups of solutions with close probabilities that correspond to additions yielding the same result: **1** $\rightarrow \{25\}$, **2** $\rightarrow \{3\}$, **3** $\rightarrow \{0, 8, 30, 12\}$, **4** $\rightarrow \{\emptyset, 17\}$, **5** $\rightarrow \{21\}$, with the branches identified according to the additional bit inversions, as shown in the first column in Eq. (5). (c) Sequential N -bit adder with the updated weights of individual FA's. The dependence on the duration of running of each individual FA is shown.

feature of results shown in Eq. (5) is the presence of several groups with the same result. These groups correspond to inverting pairs $(0, 1)$ that change arguments but preserve the sum.

The weights shown in Fig. 5(b) admit a variety of bottlenecks resulting in a poor convergence of the N -bit adder as evidenced by Fig. 3(a). Some bottlenecks can be related to the linear dependence of weights in Fig. 5(b) over $\{-1, 1\}$. As a simple “fix”, we removed some linear dependencies by increasing the magnitude of coupling in pairs (x, c_{out}) and (y, c_{out}) by 10 percent, and by scaling the weights in the k -th FA by $\sqrt{p_k}/\lfloor\sqrt{p_k}\rfloor$, where $\{p_k\}$, with $k = 1, \dots, N$, is a set of random distinct prime numbers (any set of numbers linearly independent over rationals suffices). Figure 3(b) confirms that eliminating linear dependencies drastically improves the probability of converging to correct solution. To demonstrate the performance of a cascaded circuit in the setup with eliminated mutual impact of different gates, Fig. 3(c) shows an improvement of the probability of obtaining the correct result when individual FA's are processed individually one-by-one.

Whether dynamical bottlenecks can be completely eliminated in a proper network of relaxed spins, possibly by introducing additional internal dynamical spins, ascends

282 to the question whether the isotonicity of a Boolean function is sufficient for ensuring
283 single stable equilibrium of the network. For the case of a scalar Boolean function with-
284 out internal spins, the answer is affirmative. Functions of a more complex structure
285 are a subject of ongoing research.

286 3 Discussion

287 We have demonstrated that classical dynamical systems can perform multiple univer-
288 sal computations simultaneously within a single system, without spatial or temporal
289 separation of tasks, through the mechanism of symmetry-induced parallelism, in which
290 multiple inputs and their corresponding outputs are contained within the symmetry
291 of the dynamical system. A distinguishing feature of this mechanism is that it holds
292 for nonlinear systems and therefore does not limit the character of computations. We
293 show that relaxed spin networks driven by the V_2 model enable this mechanism owing
294 to the model's dynamical properties.

295 The group of global symmetries of the V_2 model is one-parameter, which imposes
296 an ordering on the relaxed spin network phase space. As a result, simultaneous com-
297 putations arise from bit-flip chains of arguments, including the parameterization of
298 the family of evaluated functions, that are traversed by the symmetry transformation.
299 We show that simultaneous multiple evaluations correspond to the evaluated func-
300 tion being isotone along the induced argument chain. The isotonicity ensures that
301 there exists an association between input and output bits such that the variation of
302 the output bits simultaneous with the inversion of the input bits along the induced
303 chain is consistent with the function's truth table. In turn, the role of the V_2 model
304 is to dynamically solve the combinatorial problem of associating the input and out-
305 put relaxed spins so as to form a state producing a valid chain of outputs under the
306 symmetry transformation. For a simple class of Boolean functions, we show that the

isotonicity guarantees the convergence to the proper terminal state representing multiple computations. Resolving this problem for more complex Boolean functions will determine whether symmetry-induced parallelism is a deterministic feature that can be purposefully reproduced in spin representations of elaborate circuits or whether it becomes probabilistic for sufficiently complex Boolean functions.

The symmetry principle by itself does not impose inherent limitations on the character of simultaneous computations. The reliance on isotonicity may appear to be a *coincidental* consequence of the fact that the symmetry group is one-parameter. Therefore a different perspective is important: isotonicity presents a foundation for computational properties of the association between input and output bits. Such an association, in turn, is necessary for synchronously traversing various arguments and outputs by a transformation that acts locally on the network graph and affects individual relaxed spins independently. Thus, the limitations on the realization of symmetry-induced parallelism considered in the present paper originate from the nature of the symmetry transformations rather than from the symmetry group being one-parameter.

4 Methods

4.1 V_2 model

The V_2 model [8, 15, 16] governs the dynamics of N dynamical variables $\xi = (\xi_1, \dots, \xi_N) \in \mathbb{R}^N$ in a network described by the weighted adjacency matrix $\hat{A}$ with matrix elements $A_{m,n}$. Coupling between ξ_m is accounted by introducing new dynamical variables, relaxed spins, $\xi_m \rightarrow \psi_m = (\sigma_m, X_m)$, through the relation $\xi_m = \sigma_m + X_m + kP$, where $\sigma_m \in \{-1, 1\}$ is the discrete component of the relaxed spin, $X_m \in [-1, 1)$ is the continuous component, P is the period of the coupling function and k is an integer. Thus, the relaxed spin dynamics is defined on the phase space of the system of relaxed spins that has the structure shown in Fig. 4(a). When the

333 evolution carries the phase point across the boundary of the domain of the continuous
 334 component of the relaxed spin, the discrete component changes sign.

335 A transient model state $\psi = (\sigma, \mathbf{X})$ is characterized by an objective function
 336 that plays a role of a Lyapunov function for deriving the equations of motion gov-
 337 erning the evolution of the Ising machine. The objective function can be introduced
 338 in various ways: associating with σ the energy of a classical Ising model $\mathcal{H}(\sigma) =$
 339 $2^{-1} \sum_{m,n} A_{m,n} \sigma_m \sigma_n$ or the total weight of edges connecting spins with opposite ori-
 340 entations (the weight of cut of the network graph induced by partition described by
 341 the spin configuration) $\mathcal{C}(\sigma) = 4^{-1} \sum_{m,n} A_{m,n} (1 - \sigma_m \sigma_n)$. These approaches, while
 342 formally equivalent, emphasize different perspective. Therefore, to simplify different
 343 discussions, both approaches are used throughout the paper.

344 Consequently, the V_2 model is defined by the form of the objective function char-
 345 acterizing the spin configuration. For example, the relaxed cut function determining
 346 the evolution of the V_2 model can be written as

$$\mathcal{C}_{V_2}(\sigma, \mathbf{X}) = \mathcal{C}(\sigma) + \Delta \mathcal{C}_{V_2}(\sigma, \mathbf{X}), \quad (6)$$

347 where

$$\Delta \mathcal{C}_{V_2}(\sigma, \mathbf{X}) = \frac{1}{4} \sum_{m,n} A_{m,n} \sigma_m \sigma_n |X_m - X_n| \quad (7)$$

348 is the correction due to relaxation.

349 The equations of motion governing the evolution of the relaxed spins are for-
 350 mulated as ensuring that $\mathcal{C}_{V_2}(\psi) = \mathcal{C}_{V_2}(\sigma, \mathbf{X})$ does not decrease with time: $\dot{X}_m =$
 351 $2\partial \Delta \mathcal{C}_{V_2}(\sigma, \mathbf{X}) / \partial X_m$ (the factor 2 is introduced to simplify the expressions), or, more
 352 explicitly,

$$\frac{dX_m}{dt} = \sigma_m \sum_n A_{m,n} \sigma_n \operatorname{sgn}(X_m - X_n). \quad (8)$$

353 Here, $\operatorname{sgn}(X)$ is the sign function ($\operatorname{sgn}(X) = 1$, if $X > 0$, and $\operatorname{sgn}(X) = -1$, if
 354 $X < 0$) with the convention $\operatorname{sgn}(0) = 0$. Despite the discontinuous variation of both

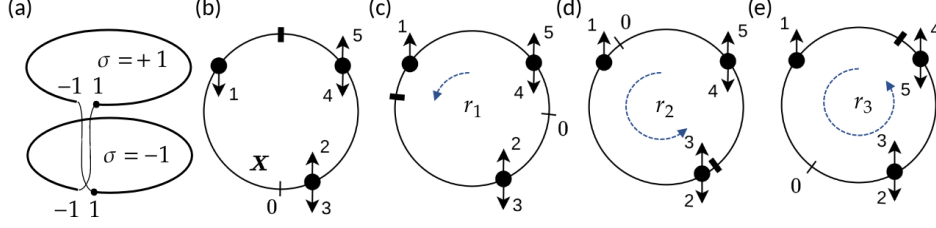


Fig. 4 Relaxed spin states. (a) The phase space of relaxed spins. Progressing across the point of contact is accompanied with inverting the discrete component. (b) A terminal state of the V_2 model on $\mathcal{K}_5$ graph starting from a random initial state has the form of three strong clusters: two containing two relaxed spins and one consisting of only one. The bold dash at the top shows the $(-1, 1)$ -boundary. Numbers near the arrows correspond to enumeration of the graph nodes. (c)-(e) Variations of the spin configuration with three counterclockwise rotations of the phase circle, or, alternatively, with the displacements of the origin by $0 < r_1 < r_2 < r_3 < 2$. The dashed curved line shows the displacement of the $(-1, 1)$ -boundary passing the strong clusters.

355 components of the relaxed spin, the relaxed cut function $\mathcal{C}_{V_2}(\psi)$ and rates dX_m/dt
 356 remain continuous during these transitions.

357 4.2 Properties enabling symmetry-induced parallelism

358 The first key feature of the V_2 model is the structure of the terminal states: generally,
 359 relaxed spins form groups (*strong clusters*) with coinciding continuous components
 360 (see Theorem 1 in [8]). Figure 4(b) shows the clustered structure of the terminal state
 361 for the example of $\mathcal{K}_5$, complete graph with five nodes.

362 The relaxed cut function $\mathcal{C}_{V_2}(\psi)$ is invariant with respect to homogeneous trans-
 363 lations of all continuous components: $\mathbf{X} \rightarrow \mathbf{X} + r\mathbf{1}$, where $\mathbf{1} = (1, 1, \dots, 1)$, or,
 364 element-wise, $X_m \rightarrow X_m + r$. Alternatively, this invariance can be interpreted as
 365 the symmetry with respect to rotation of the phase circle (displacement of the ori-
 366 gin). Because of the continuity of $\mathcal{C}_{V_2}(\psi)$, this rotation can be extended to arbitrary
 367 real r , including those that lead to strong clusters crossing from one phase circle to
 368 another in the representation shown in Fig. 4(a) or to traverse the $(-1, 1)$ -boundary
 369 in the representation as in Fig. 4(b). As a result, the symmetry transformation
 370 leads to a variation of the discrete component of the relaxed spin configuration.
 371 Figures 4(c)-(e) show an example of such state transformations with the original

372 discrete component $\sigma = (-1, 1, -1, -1, 1)$ [Fig. 4(b)]. After the first rotation, the
 373 configuration becomes $\sigma = (1, 1, -1, -1, 1)$ [Fig. 4(c)]. After the second rotation, we
 374 obtain $\sigma = (1, -1, 1, -1, 1)$ [Fig. 4(d)]. Finally, the third rotation produces the full
 375 inversion of the original state $\sigma = (1, -1, 1, 1, -1)$ [Fig. 4(exs)].

376 In view of these variations, the critical feature of the V_2 model is that, if the
 377 system Hamiltonian is strictly quadratic in spin variables, the correction to the relaxed
 378 cut function vanishes in the terminal state, $\Delta\mathcal{C}_{V_2}(\psi) = 0$. Hence, all the discrete
 379 spin configurations obtained while applying symmetry transformations produce the
 380 same value of the (discrete) cut function $\mathcal{C}(\sigma)$. This property trivializes rounding the
 381 terminal states of the V_2 model: regardless of the choice of the phase circle origin, the
 382 obtained spin configurations correspond to the same value of the objective function
 383 (Theorem 3 in [8]). In particular, if one configuration produces the maximum value
 384 of $\mathcal{C}(\sigma)$, then all of them do. It is not difficult to see that this property significantly
 385 relies on the strong clustered structure of the terminal state. Therefore, in regularized
 386 V_2 models, with $\text{sgn}(X)$ in the right-hand-side of the equations of motion replaced by
 387 a continuous function deviating from $\text{sgn}(X)$ in an interval containing the origin, the
 388 obtained spin configuration may depend on whether the displaced $(-1, 1)$ -boundary
 389 falls inside the regularization interval for some node pairs. These situations can be
 390 consistently incorporated into the theoretical framework with the help of the notion of
 391 *weak clusters* [16]. However, such an elaborated approach is excessive for the purposes
 392 of the present paper.

393 4.3 Spin representations of Boolean functions

394 The V_2 model is employed by representing evaluating a Boolean function as finding
 395 the ground state (the maximum cut partition) of the respective graph. The general
 396 principle of such representations is to designate some spins in the network as input bits,
 397 $\sigma^{(\text{in})}$, up to the relation between binary $\{0, 1\}$ and spin $\{-1, 1\}$ variables, while some

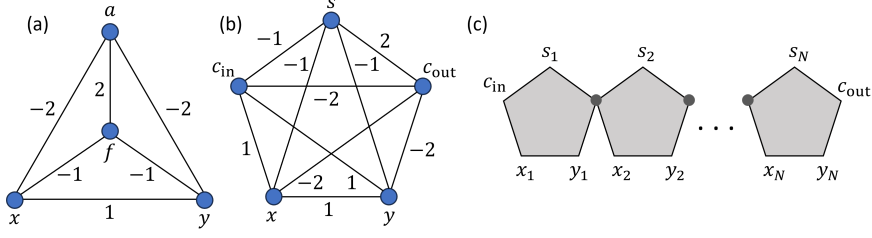


Fig. 5 Spin representations of the logical gates used for illustration of symmetry-induced parallelism. For shown weights, values obtained in [14] are used. (a) AND/OR gate. The operation regime is determined by σ_f , the value of the fixed auxiliary spin: $\sigma_f = 1$ and $\sigma_f = -1$ correspond to AND and OR, respectively. (b) Full-adder, FA. (c) N -bit adder composed of N FA's. The carry-out spin of the k -th FA is identified with carry-in spin of the $(k + 1)$ -th FA.

spins play the role of the output bits, $\sigma^{(\text{out})}$.¹ Consequently, the Ising Hamiltonian $\mathcal{H}(\sigma)$ represents a vector-valued Boolean function $f : \{-1, 1\}^N \rightarrow \{-1, 1\}^M$, if in the ground state given the state of input spins $\sigma^{(\text{in})}$, the state of the output spins is $\sigma^{(\text{out})} = f(\sigma^{(\text{in})})$. Writing the problem of finding the Ising model ground state with fixed certain spins as

$$\sigma_* = \arg \min_{\sigma | \sigma^{(\text{in})}} \mathcal{H}(\sigma), \quad (9)$$

the value of the Boolean function is read as the state of the output spins: $f(\sigma^{(\text{in})}) = \sigma_* |_{\sigma^{(\text{out})}}$. Equivalently, a spin representation of a Boolean function can be regarded as obtaining the maximum cut partition given the assignments of the input nodes: $\sigma_* = \arg \max_{\sigma | \sigma^{(\text{in})}} \mathcal{C}(\sigma)$.

Such representations were extensively studied in the literature [14, 17–21]. As the main text demonstrates, symmetry-induced parallelism poses its own set of requirements. However, as a starting point, it is constructive to use already studied representations. Figures 5(a, b) show the spin representations (complete graphs with properly chosen weights) of digital gates that we use to illustrate symmetry-induced parallelism: AND and 3-bit full adder (FA), developed in [14]. The respective Boolean functions can be written in terms of their arguments as $\text{AND}(x, y) = a$ and $\text{FA}(x, y, c_{\text{in}}) = \langle s, c_{\text{out}} \rangle$, where $a = x \wedge y$, $s = x \oplus y \oplus c_{\text{in}}$ is the sum bit and

¹We adopt the convention associating spin values +1 and -1 with True and False, respectively.

415 $c_{\text{out}} = \text{MAJ}(x, y, c_{\text{in}})$ is the carry-out bit. Here, $\oplus$ denotes XOR (or the sum modulo 2
 416 in the $\{0, 1\}$ representation for Boolean variables), and MAJ is the majority function
 417 (returns True, if and only if the most arguments are True). Taking into account these
 418 notations, the Hamiltonians describing the spin representations of these gates are

$$\begin{aligned}\mathcal{H}_{\text{AND}}(\boldsymbol{\sigma}) &= \sigma_x \sigma_y - 2\sigma_x \sigma_a - 2\sigma_y \sigma_a - \sigma_x \sigma_f - \sigma_y \sigma_f + 2\sigma_a \sigma_f, \\ \mathcal{H}_{\text{FA}}(\boldsymbol{\sigma}) &= 2\sigma_o \sigma_s - 2\sigma_o \sigma_x - 2\sigma_o \sigma_y - 2\sigma_o \sigma_i \\ &\quad - \sigma_s \sigma_x - \sigma_s \sigma_y - \sigma_s \sigma_i + \sigma_x \sigma_y + \sigma_x \sigma_i + \sigma_y \sigma_i,\end{aligned}\tag{10}$$

419 where $\sigma_f \equiv 1$ is a fixed auxiliary spin, and σ_i and σ_o are spins representing carry-in
 420 and carry-out, respectively.

421 The fixed auxiliary spin is introduced in the spin representations of AND to ensure
 422 that $\mathcal{H}_{\text{AND}}(\boldsymbol{\sigma})$ is strictly quadratic. This reflects the fact that the truth table of AND,
 423 in contrast to FA, is not invariant with respect to inversion of all spins, which yields
 424 OR function. Since an unconstrained rotation of the phase circle results in inversion of
 425 the auxiliary spin, the respective gate is called AND/OR in the main text. It is worth
 426 noting that substituting $\sigma_a \rightarrow -\sigma_a$ in $\mathcal{H}_{\text{AND}}(\boldsymbol{\sigma})$ yields the Hamiltonian corresponding
 427 to the NAND/NOR gate.

428 Constructing spin representations of complex Boolean functions out of more
 429 elementary, say, logical gates, poses little constraints on spin representations of indi-
 430 vidual sub-functions. It is sufficient that the cut values obtained whenever the gate
 431 equation is satisfied, $\boldsymbol{\sigma}^{(\text{out})} = f_i(\boldsymbol{\sigma}^{(\text{in})})$, where i is the gate number, are the same
 432 $\max_{\boldsymbol{\sigma}|\boldsymbol{\sigma}^{(\text{in})}} \mathcal{C}_i(\boldsymbol{\sigma}) = C$ for all gates, while the cut values corresponding to unsatisfied
 433 gate equations are strictly smaller. Then, for any consistent circuit, that is such that
 434 there exists an assignment of Boolean values to all wires satisfying simultaneously
 435 every gate equation, using the spin representations of individual gates will produce a
 436 valid spin representation $\mathcal{G}$ of the whole circuit. Indeed, let us consider a maximum
 437 cut of this graph given the values of input parameters $\overline{C}_{\mathcal{G}} = \mathcal{C}_{\mathcal{G}}(\boldsymbol{\sigma}_*)$. Obviously, the

cut of any induced subgraph evaluated for the respective subset of σ_* cannot exceed the maximum cut of this subgraph. Since the spin representations of individual gates do not share edges, we have $\overline{C}_{\mathcal{G}} \leq NC$, where N is number of gates in the circuit. On the other hand, since the circuit is consistent, one can assign the input and output spins of each gate according to the Boolean states of the circuit wires given the input. Such an assignment will maximize the cut for each gate representing graph and, hence, $\overline{C}_{\mathcal{G}} \geq NC$. Thus, we have $\overline{C}_{\mathcal{G}} = NC$, and σ_* is the spin assignment satisfying all gate equations in the circuit with the circuit output spins corresponding to the value of the represented Boolean function given its input.

It must be noted that the existence of broken chains prevents translating this argument straightforwardly to the property of symmetry-induced parallelism.

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500 Code Availability

501 The code used for preparing Fig. 3 is available at [https://github.com/merement/](https://github.com/merement/Dice/tree/main/cases/s_parallelism)
502 [Dice/tree/main/cases/s_parallelism](https://github.com/merement/Dice/tree/main/cases/s_parallelism). For simulating the R-DIM, library available at
503 <https://github.com/merement/Dice> is used.

504 **Data Availability**

505 Simulation data presented in Fig. 3 is available at [https://github.com/merement/](https://github.com/merement/Dice/tree/main/cases/s_parallelism/data/sims)
506 [Dice/tree/main/cases/s_parallelism/data/sims](https://github.com/merement/Dice/tree/main/cases/s_parallelism/data/sims).

507 **Competing Interest**

508 The authors declare no competing interests.

509 **Author Contribution**

510 Conceptualization: M.E.; Methodology: M.E.; Visualization: M.E.; Discussion: M.E.,
511 P.M.; Funding acquisition: M.E., P.M.; Supervision: P.M.; Writing: M.E.; Revision:
512 M.E., P.M.